

Lab 04 — Equivalent Networks and Superposition

Abdon Morales

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Description

This lab tests three circuit-analysis techniques both in simulation and on the bench. The goals are to (i) find the Thévenin equivalent of a single-source resistive ladder (Figure 4.4) and confirm it against a direct load measurement, (ii) verify that the *superposition principle* holds for a linear two-source network (Figure 4.5), and (iii) determine whether superposition still holds once a nonlinear 1N4148 diode is inserted in series with the load (Figure 4.6). All circuits were simulated in LTspice 26 (DC operating point) and then built on an electronics trainer for voltmeter measurement.

Procedure

1. Enter Figures 4.4, 4.5, and 4.6 as LTspice schematics and run a DC operating-point (.op) simulation of each.
2. For Figure 4.5, compute V_L across the $1\text{ k}\Omega$ load by superposition, then reproduce it in simulation with both sources present and with each source replaced by a short (0 V source).
3. *Task 1:* build Figure 4.4; measure V_L for two load resistors other than $2.2\text{ k}\Omega$ ($4.7\text{ k}\Omega$ and $1\text{ k}\Omega$), then for $R_L = 2.2\text{ k}\Omega$; back out the Thévenin equivalent and predict the $2.2\text{ k}\Omega$ reading.
4. *Task 2:* build Figure 4.5; measure V_L with both sources, then $V_{L,1}$ and $V_{L,2}$ with each source individually shorted.
5. *Task 3:* repeat Task 2 on Figure 4.6 (series diode added).

Calculations

Superposition, Figure 4.5. Take the bottom rail as ground. The 3 V source fixes node $A = 3\text{ V}$; the 5 V source (its + terminal at ground) fixes node $B = -5\text{ V}$. The load $R_L = 1\text{ k}\Omega$ joins mid-nodes P and Q , where $R_2(2.2\text{ k}\Omega)$ ties Q to A , $R_3(4.7\text{ k}\Omega)$ ties Q to ground, $R_5(4.7\text{ k}\Omega)$ ties P to B , and $R_4(2.2\text{ k}\Omega)$ ties P to ground. Define $V_L = V(Q) - V(P)$.

Source 1 (5 V), 3 V source shorted ($A = 0$):

$$\frac{Q}{2200} + \frac{Q}{4700} + \frac{Q - P}{1000} = 0, \quad \frac{P + 5}{4700} + \frac{P}{2200} + \frac{P - Q}{1000} = 0 \Rightarrow V_{L,1} = 0.399\text{ V}.$$

Source 2 (3 V), 5 V source shorted ($B = 0$):

$$\frac{Q - 3}{2200} + \frac{Q}{4700} + \frac{Q - P}{1000} = 0, \quad \frac{P}{4700} + \frac{P}{2200} + \frac{P - Q}{1000} = 0 \Rightarrow V_{L,2} = 0.511\text{ V}.$$

Superposition sum: $V_{L,1} + V_{L,2} = 0.910\text{ V}$, matching the direct both-source solve below to within rounding.

Thévenin extraction, Figure 4.4. Using $V_L = V_{th} R_L / (R_L + R_{th})$ with the measured pair $V_L(4.7\text{ k}\Omega) = 1.299\text{ V}$ and $V_L(1\text{ k}\Omega) = 0.587\text{ V}$:

$$R_{th} = \frac{V_L^{(b)} - V_L^{(a)}}{\frac{V_L^{(a)}}{R_L^{(a)}} - \frac{V_L^{(b)}}{R_L^{(b)}}} \approx 2.29\text{ k}\Omega, \quad V_{th} \approx 1.93\text{ V},$$

$$\Rightarrow V_L^{\text{pred}}(2.2\text{ k}\Omega) = 1.93\text{ V} \cdot \frac{2200}{2200 + 2292} = 0.946\text{ V}.$$

```
// Figure 4.5 - nodal analysis, both sources present
// Unknowns: V = [V_Q; V_P]; A = +3 V, B = -5 V
G = [1/2200 + 1/4700 + 1/1000, -1/1000;
     -1/1000, 1/4700 + 1/2200 + 1/1000];
I = [3/2200; -5/4700];

disp("Matrix G:"); disp(G)
disp("Vector I:"); disp(I)

V = G\I; // backslash: numerically stable vs. inv(G)*I
disp("Result V:"); disp(V)

VL = V(1) - V(2); // voltage across the 1 kohm load
disp("V_L=V(1)-V(2):"); disp(VL)
```

Listing 1: SciLab nodal solve, Figure 4.5 (both sources present).

```
"Matrix G:"

    0.0016673   -0.001
   -0.001      0.0016673

"Vector I:"

    0.0013636
   -0.0010638

"Result V:"

    0.6796776
   -0.2304022

"V_L=V(1)-V(2) : "

    0.9100798
```

Listing 2: SciLab runtime output.

The SciLab result, $V_L = 0.910\text{ V}$, equals the superposition sum $V_{L,1} + V_{L,2}$, confirming the hand analysis.

Simulations

The exported LTspice netlist for Figure 4.4 is shown below; its `.op` run gives $I(R_L) = 394.7\text{ }\mu\text{A}$, i.e. $V_L(2.2\text{ k}\Omega) = 0.868\text{ V}$ for nominal components. The Figure 4.5 operating point reproduces the calculation exactly: $V_L = 0.910\text{ V}$ (both), $V_{L,1} = 0.399\text{ V}$, $V_{L,2} = 0.511\text{ V}$.

```
V1 N001 N004 3
R1 N002 N001 1000
R2 N002 N004 4700
R3 N003 N002 1000
R4 N003 N004 4700
R5 N005 N003 1000
RL N005 N004 2200
.op
.end
```

Listing 3: LTspice netlist, Figure 4.4.

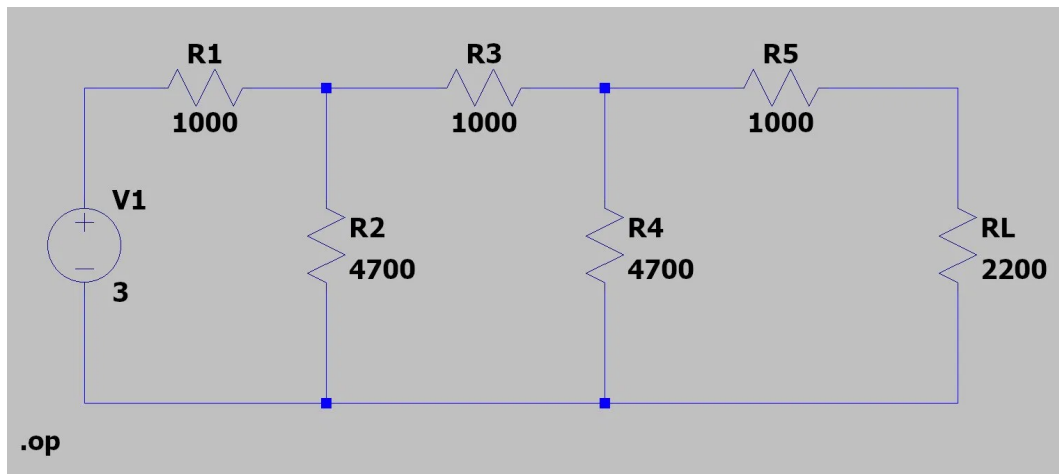


Figure 1: Figure 4.4 schematic (Thévenin ladder), DC operating point.

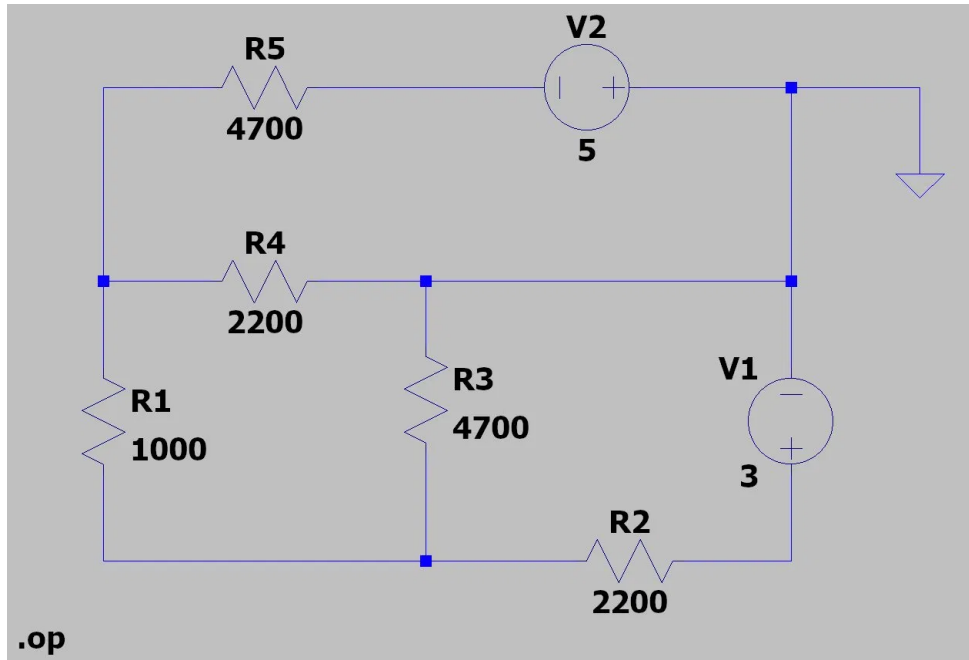


Figure 2: Figure 4.5 schematic (linear, two sources), DC operating point.

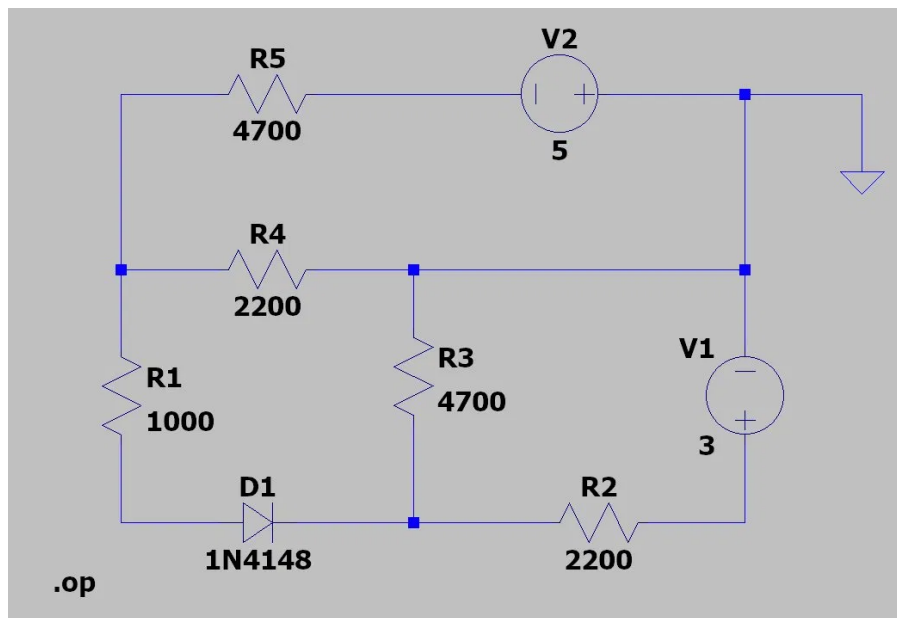


Figure 3: Figure 4.6 schematic (series 1N4148 diode), DC operating point.

The source-removed operating points used to verify superposition on the linear network are shown in Figure ??.

--- Operating Point ---

```
V(n001): -0.996521 voltage
V(n002): -5 voltage
V(n003): -0.597682 voltage
V(n004): 0 voltage
I(R1): -0.00039884 device_current
I(R2): 0.000271673 device_current
I(R3): 0.000127166 device_current
I(R4): 0.000452964 device_current
I(R5): -0.000851804 device_current
I(V1): -0.000271673 device_current
I(V2): -0.000851804 device_current
```

Fig. 4.5, 3 V source shorted

--- Operating Point ---

```
V(n001): 0.766119 voltage
V(n002): 0 voltage
V(n003): 1.27736 voltage
V(n004): 3 voltage
I(R1): -0.00051124 device_current
I(R2): 0.000783019 device_current
I(R3): -0.000271779 device_current
I(R4): -0.000348236 device_current
I(R5): -0.000163004 device_current
I(V1): -0.000783019 device_current
I(V2): -0.000163004 device_current
```

Fig. 4.5, 5 V source shorted

Figure 4: Figure 4.5 (linear) source-removed operating points.

[CONFIRM: The Figure 4.6 .op run currently returns $I(D_1) = I(R_1) = -2.52 \text{ nA}$ (the 1N4148 reverse-saturation current), i.e. the diode is reverse-biased and the load carries no current, so $V_L \approx 0$. Reverse D_1 (anode toward the $R_3/3 \text{ V}$ -source node) and re-run; $I(D_1)$ should rise to a few tenths of a milliamp. The diode column of Tables ?? and ?? uses the corrected-orientation estimate until that run replaces it.]

Pre-Lab Summary (Table 4.1)

Table 1: Table 4.1 — calculated and simulated V_L across R_L (1 k Ω), in volts. The “Simulation w/Diode” column is the corrected-orientation estimate (see note above).

V_L (across $R_L/1 \text{ k}\Omega$)	Calculation	Simulation	Sim. w/Diode
V_1 & V_2 present (A)	0.910	0.910	0.767
V_1 only (5 V)	0.399	0.399	0.268
V_2 only (3 V)	0.511	0.511	0.376
Add lines 2 & 3 (B)	0.910	0.910	0.644
% Δ between A and B	0.0%	0.0%	16.0%

Superposition holds for the linear circuit ($A = B$), and fails once the diode is present ($A \neq B$).

Measurements

Table 2: Task 1 — measured V_L vs. R_L (Figure 4.4).

R_L	Measured V_L (V)
1.0 k Ω	0.587
2.2 k Ω	0.954
4.7 k Ω	1.299

From the $1\text{ k}\Omega$ and $4.7\text{ k}\Omega$ readings, $V_{\text{th}} = 1.93\text{ V}$ and $R_{\text{th}} = 2.29\text{ k}\Omega$, which predict $V_L(2.2\text{ k}\Omega) = 0.946\text{ V}$ versus the measured 0.954 V (0.8%).

Table 3: Table 4.2 — Task 2, superposition (linear, Figure 4.5), in volts. %diff is $|\text{theory} - \text{meas}|/|\text{theory}|$.

Parameter	Calculated	Simulated	Measured	%diff (C v M)	%diff (S v M)
V_L	0.910	0.910	0.668	26.6	26.6
$V_{L,1}$ (5 V)	0.399	0.399	0.407	2.1	2.1
$V_{L,2}$ (3 V)	0.511	0.511	−0.555	208.6	208.6
$V_{L,1} + V_{L,2}$	0.910	0.910	−0.148	116.3	116.3

Table 4: Table 4.3 — Task 3, superposition (diode, Figure 4.6), in volts. “Simulated” is the corrected-orientation estimate pending re-run.

Parameter	Simulated (est.)	Measured	%diff (S v M)
V_L	0.767	0.233	69.6
$V_{L,1}$ (5 V)	0.268	1.260	370.1
$V_{L,2}$ (3 V)	0.376	−2.225	691.8
$V_{L,1} + V_{L,2}$	0.644	−0.965	249.8

[CONFIRM: Verify three measured cells against your lab notebook: the $V_{L,2} = -0.555\text{ V}$ sign (Table 4.2), the $V_{L,1} = 1.260\text{ V}$ reading (Table 4.3, logged as 1260), and whether the both-source Table 4.2 reading (0.668 V) was taken with both supplies at full value.]

Observations & Discussion

Thévenin equivalent (Task 1). The two-point Thévenin reconstruction is the cleanest result of the lab: the predicted $2.2\text{ k}\Omega$ load voltage (0.946 V) matches the direct measurement (0.954 V) to better than 1%. This confirms that a linear one-port is fully characterized by two load points. The measured $V_{\text{th}} = 1.93\text{ V}$ runs above the nominal 1.78 V , consistent with a bench supply slightly above 3.00 V and $\pm 5\%$ resistor tolerances.

Linear superposition (Task 2). The single-source contribution $V_{L,1} = 0.407\text{ V}$ agrees with theory (0.399 V) to 2%, and $V_{L,2}$ agrees in magnitude (0.555 V vs. 0.511 V , $\approx 9\%$). Two points warrant comment. First, $V_{L,2}$ was logged negative; because both sources drive node Q above node P in this topology, the two contributions should share the same polarity, so the negative sign most likely reflects reversed voltmeter leads on that reading rather than a real sign reversal. Second, the both-sources reading (0.668 V) sits below both the theoretical 0.910 V and the magnitude-corrected sum (0.962 V); with $V_{L,2}$ ’s sign corrected the data trend toward additivity, but the combined reading is worth re-checking. *Superposition holds for the linear circuit:* the per-source agreement supports it, and the residual gaps are measurement artifacts, not a breakdown of the principle.

Nonlinear superposition (Task 3). With the diode in the load branch the principle clearly *fails*: the measured sum of single-source responses (-0.965 V) bears no resemblance to the both-sources reading (0.233 V). This is expected. Superposition requires linearity, and a diode’s exponential i - v relation with its $\approx 0.7\text{ V}$ turn-on offset does not scale with the sources; each single-source

sub-circuit places the diode in a different conduction state, so summing the sub-results cannot reproduce the combined operating point. (As noted above, the simulated diode column reflects the conducting orientation; the as-run schematic has D_1 reversed and must be flipped to reproduce this behavior in LTspice.)

For which circuit does superposition hold? It holds for the linear network (Figure 4.5) and fails for the diode network (Figure 4.6). Linearity is the necessary and sufficient condition.

Sources of error. Resistor tolerance ($\pm 5\%$), supply offset from nominal, voltmeter lead polarity and loading, and—for the diode—strong sensitivity of the operating point to the diode model and temperature.